

DD_SUB-WF-PAUSE-01-FLOW-WIK - WIK Pause BPMN

Developer implementation guide: DD_SUB-WF-PAUSE-01-FLOW-WIK-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This DD defines the WIK operator BPMN for voluntary subscription pause.

Process key:

sub-pause-wik

This BPMN implements the parent pause contract from DD_SUB-WF-PAUSE-01-Subscription_Pause-v1.0.md for WIK.

The WIK pause flow is normally no-fee in v1.0:

validate -> PENDING_PAUSE -> compute billing window -> emit billing intent -> optional fee payment -> suspend service -> c

The optional pay-first gate exists in the BPMN, but it is dormant for the seeded WIK v1.0 pause reasons because no seeded reason has a pause fee.

2. Ownership

This DD owns:

- WIK-specific pause BPMN process key.
- WIK happy-path BPMN sequence.
- WIK reason catalog seed values.
- WIK duration behavior.
- WIK optional payment gate behavior.
- WIK fulfillment suspension behavior.
- WIK worker-topic composition.
- WIK message catches.
- WIK use of rules.subscription.pause.wik.
- WIK pause examples.

This DD does not own:

- Generic pause API contract and schemas. See parent pause DD.
- Shared operation framework. See SUB-WF-FRAMEWORK-01.
- Worker contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- Worker implementation code.
- SUB-LM, BIL, FUL, NOT, scheduler internals.
- Resume implementation. See SUB-WF-RESUME-01.

3. Required Runtime Configuration

3.1 Framework Operation Config

What this table is for:

subscription_operation_config tells the subscription workflow framework which BPMN process to start for an operator and operation kind.

When to use it:

Use it when the API receives a PAUSE request and needs to resolve the WIK BPMN process key.

Required row:

```

INSERT INTO subscription_operation_config (
    operator_code,
    operation_kind,
    default_bpmn_process_key,
    operation_timeout_seconds,
    feature_flags,
    enabled,
    updated_by
)
VALUES (
    'WIK',
    'PAUSE',
    'sub-pause-wik',
    120,
    '{}',
    TRUE,
    'migration'
);

```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	This row applies to WIK.
operation_kind	Subscription operation.	PAUSE	This row is used for pause requests.
default_bpmn_process_key	Camunda process key to start.	sub-pause-wik	Start the WIK pause BPMN.
operation_timeout_seconds	Framework operation timeout.	120	Pause should normally finish quickly unless waiting on payment or fulfillment.
feature_flags	Operator-specific flags.	{}	No WIK pause flow flags in v1.0.
enabled	Runtime enablement.	TRUE	API may start WIK pause.
updated_by	Audit actor.	migration	Seeded by migration.

3.2 WIK Pause Config

What this table is for:

subscription_pause_config stores operator-level pause policy, such as whether self-service pause is allowed and how far in the future a resume can be scheduled.

When to use it:

Load it before Drools validation and pass it as a fact into rules.subscription.pause.wik.

Required row:

```

INSERT INTO subscription_pause_config (
    operator_code,
    customer_self_service_enabled,
    scheduled_pause_enabled,
    max_future_scheduled_resume_days,
    min_pause_hours,
    customer_notification_enabled,
    updated_by
)
VALUES (
    'WIK',
    TRUE,
    TRUE,
    90,
    24,
    TRUE,
    'migration'
);

```

Column meaning:

Column	Purpose	Sample	Meaning
operator_code	Operator scope.	WIK	Policy applies to WIK.
customer_self_service_enabled	Allows customer portal pause.	TRUE	Customers may request eligible pause reasons themselves.
scheduled_pause_enabled	Allows scheduled or fixed-duration pause.	TRUE	WIK supports fixed days and scheduled resume.

Column	Purpose	Sample	Meaning
max_future_scheduled_resume_days	Maximum future resume horizon.	90	Resume cannot be scheduled more than 90 days out.
min_pause_hours	Minimum pause duration.	24	Pause must last at least one day.
customer_notification_enabled	Enables NOT-01 pause notifications.	TRUE	Send customer notification after success.
updated_by	Audit actor.	migration	Seeded by migration.

3.3 WIK Pause Reasons

What this table is for:

pause_reason is the pause reason catalog. It defines which reasons customers can use, which reasons are admin-only, duration bounds, billing impact, and whether network service must be suspended.

When to use it:

Load the requested reason before Drools validation, snapshot it into the operation variables, and store the snapshot in subscription_pause_history after the pause succeeds.

Required seed data:

code	customer_self_service_eligible	admin_only	max_duration_days	open_ended_allowed	pauses_bill_cycle	suspends_fulfillment
CUSTOMER_VACATION	TRUE	FALSE	90	FALSE	TRUE	TRUE
CUSTOMER_TEMPORARY_AWAY	TRUE	FALSE	30	FALSE	TRUE	TRUE
CUSTOMER_FINANCIAL_HARDSHIP	FALSE	FALSE	60	FALSE	TRUE	TRUE
ADMIN_MAINTENANCE	FALSE	TRUE	NULL	TRUE	FALSE	TRUE
ADMIN_COMPLIANCE_HOLD	FALSE	TRUE	NULL	TRUE	FALSE	TRUE

Sample row meaning:

CUSTOMER_VACATION means a customer, call-center agent, or subscription operator can pause an active subscription for up to 90 days. The pause stops bill-cycle accrual and suspends network service.

4. BPMN Start Variables

The shared framework starts Camunda with these variables.

Variable	Purpose	Example
operationId	Links BPMN to subscription_operation.	subop_01HZ_KAMAU_PAUSE
operatorCode	Operator scope.	WIK
subscriptionId	Subscription being paused.	sub_01HZ_KAMAU_FIBER
operationKind	Always PAUSE.	PAUSE
customerId	Customer reference from SUB-LM.	cust_01HZ_KAMAU
pauseTrigger	Why pause was requested.	CUSTOMER_REQUESTED_PAUSE
pauseReasonCode	Requested pause reason.	CUSTOMER_VACATION
durationMode	Resolved duration mode.	FIXED_DAYS
durationDays	Fixed duration days, if supplied.	31
resumeScheduledAt	Resolved resume timestamp, if scheduled.	2026-04-15T10:00:00Z
reasonCatalogSnapshot	Snapshot of the pause reason row.	{"pauses_bill_cycle":true}
pauseFeeApplicable	Whether payment gate should run.	false
initiatingActorUserId	Actor user/service.	cc-agent-wik-007
initiatingActorRole	Actor role.	CALL_CENTER
initiatingWorkflowKind	Upstream workflow.	BO_UI

Variable	Purpose	Example
initiatingWorkflowRef	Upstream reference.	cc-ticket-2026-03-15-3301

5. Worker Topics Used

These are logical Camunda external-task topics, not separate deployments. One worker app may subscribe to many topics, but logs, retries, metrics, and contracts must remain visible per topic.

Worker topic	Why WIK pause uses it
drools-decide	Re-check pause policy, reason, role, active status, and duration.
sub-lm-put-pending-status	Move subscription to PENDING_PAUSE.
sub-lm-compute-cycle-window	Compute billing-cycle context for BIL-01.
bil01-emit-intent	Ask BIL-01 to evaluate PAUSE_INTENT.
bil01-charge-pre-authorize	Optional payment gate when pause fee applies.
bil01-confirm-charge	Optional confirmation after payment message.
ful-call-restrict	Ask FUL-04 to suspend service using SUSPEND_SERVICE.
sub-lm-put-active-restrictions	Clear active restrictions after service suspension.
sub-lm-put-master-fields	Write status audit fields and pause-history row.
sub-lm-commit-state	Commit subscription to SUSPENDED.
kafka-emit	Write pause success/failure event to outbox.
notification-send	Send pause notification without failing BPMN on notification failure.
subscription-operation-finalize	Set operation terminal state.
bil01-revert-pending-charge	Compensation for pending pause fee charge.
sub-lm-revert-to-prior	Compensation to return subscription to ACTIVE.

6. Happy Path BPMN

```

Start
-> update operation state VALIDATING
-> drools-decide
-> gateway: eligible and duration valid?
    no -> pause rejected path
    yes
-> update operation state PENDING_STATE_FLIP
-> sub-lm-put-pending-status
-> sub-lm-compute-cycle-window
-> update operation state BILLING_CALL
-> bil01-emit-intent
-> gateway: pauseFeeApplicable?
    yes -> bil01-charge-pre-authorize -> wait payment-confirmed -> bil01-confirm-charge
    no -> continue
-> update operation state FULFILLMENT_CALL
-> ful-call-restrict
-> update operation state AWAITING_FULFILLMENT_RESPONSE
-> wait fulfillment-completed
-> sub-lm-put-active-restrictions CLEAR_ALL
-> update operation state COMMITTING_FINAL_STATE
-> sub-lm-put-master-fields
-> sub-lm-commit-state
-> update operation state EMITTING_EVENT
-> kafka-emit SubscriptionPaused
-> notification-send
-> subscription-operation-finalize COMPLETED
-> End

```

7. Step Details

7.1 drools-decide

Inputs:

```

{
  "kjarPackage": "rules.subscription.pause.wik",
  "requestedDecisions": [
    "validate-preconditions",

```

```

        "resolve-duration-mode",
        "compute-pause-fee-applicability"
    ],
    "factsJson": "{...pause facts...}"
}

```

Expected output for WIK vacation pause:

```

{
  "decisionsJson": {
    "eligible": true,
    "durationMode": "FIXED_DAYS",
    "durationDays": 31,
    "resumeScheduledAt": "2026-04-15T10:00:00Z",
    "durationValid": true,
    "pauseFeeApplicable": false,
    "reasonCatalogSnapshot": {
      "code": "CUSTOMER_VACATION",
      "pauses_bill_cycle": true,
      "suspends_fulfillment": true,
      "max_duration_days": 90
    }
  }
}

```

If eligible = false or durationValid = false, route to pause rejection.

7.2 sub-lm-put-pending-status

Inputs:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "toStatus": "PENDING_PAUSE",
  "currentTransitionType": "PAUSE",
  "currentTransitionReasonCode": "CUSTOMER_VACATION"
}

```

Output consumed later:

```

{
  "priorStatusSnapshot": {
    "statusCode": "ACTIVE",
    "currentTransitionType": null,
    "currentTransitionReasonCode": null
  }
}

```

7.3 bil01-emit-intent

Inputs:

```

{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intentKind": "PAUSE_INTENT",
  "intentContext": {
    "pauseReasonCode": "CUSTOMER_VACATION",
    "durationMode": "FIXED_DAYS",
    "durationDays": 31,
    "resumeScheduledAt": "2026-04-15T10:00:00Z",
    "pausesBillCycle": true,
    "suspendsFulfillment": true,
    "originIntent": "CUSTOMER_REQUESTED_PAUSE"
  },
  "cycleWindow": {
    "cycleStart": "2026-03-01",
    "cycleEnd": "2026-03-31"
  }
}

```

Expected WIK v1.0 result for seeded vacation reason:

```

{
  "intentDecisionId": "bidec_01HZ_KAMAU_PAUSE",
  "lineItems": [],
  "totals": {
    "net": 0,
    "currency": "KES"
  }
}

```

7.4 Optional Payment Gate

The payment gate runs only if Drools returns:

```
{
  "pauseFeeApplicable": true
}
```

For WIK seeded v1.0 reasons this is normally false, so BPMN skips:

```
bil01-charge-pre-authorize -> payment-confirmed -> bil01-confirm-charge
```

Keep the gateway in the BPMN because future WIK pause reasons may carry a pause fee.

7.5 ful-call-restrict

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "intent": "SUSPEND_SERVICE",
  "reasonCategory": "CUSTOMER_VACATION",
  "suspendsFulfillment": true,
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_PAUSE:pause"
}
```

FUL-04 suspends the customer service, for example voice profile and broadband authentication. The BPMN waits for fulfillment-completed or fulfillment-failed correlated by fulfillmentCorrelationId.

7.6 sub-lm-put-master-fields

This step writes final pause metadata before the state is committed.

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "fieldsJson": {
    "last_status_changed_at": "2026-03-15T10:00:05Z"
  },
  "pauseHistoryRow": {
    "origin_intent": "CUSTOMER_REQUESTED_PAUSE",
    "system_managed": false,
    "pause_reason_code": "CUSTOMER_VACATION",
    "pause_reason_snapshot": {
      "code": "CUSTOMER_VACATION",
      "pauses_bill_cycle": true,
      "suspends_fulfillment": true,
      "max_duration_days": 90
    },
    "duration_mode": "FIXED_DAYS",
    "duration_days": 31,
    "resume_scheduled_at": "2026-04-15T10:00:00Z",
    "pause_fee_billable_event_id": null,
    "suspended_at": "2026-03-15T10:00:05Z",
    "actual_resume_at": null,
    "pause_correlation_id": "subop_01HZ_KAMAU_PAUSE",
    "pause_actor_user_id": "cc-agent-wik-007",
    "pause_actor_role": "CALL_CENTER",
    "notes": "Customer travelling to Mombasa for holiday season"
  }
}
```

The worker implementation may call SUB-LM and pause workflow persistence behind one topic. Do not write this row with ad hoc SQL inside the BPMN worker if the project has a dedicated repository/service for pause history.

7.7 sub-lm-commit-state

Inputs:

```
{
  "subscriptionId": "sub_01HZ_KAMAU_FIBER",
  "expectedFromStatus": "PENDING_PAUSE",
  "toStatus": "SUSPENDED"
}
```

Emit success only after this commit succeeds.

8. Message Catches

Message	Correlation key	Used when
payment-confirmed	billableEventId	Only if a pause fee was pre-authorized.
fulfillment-completed	fulfillmentCorrelationId	FUL-04 successfully suspended service.
fulfillment-failed	fulfillmentCorrelationId	FUL-04 failed or rejected suspension.

Payment wait has a 7-day boundary timer. On timeout, BPMN compensates and finalizes the operation as FAILED.

9. Drools Package

Required package:

```
rules.subscription.pause.wik
```

Required decisions:

Decision	Purpose
validate-preconditions	Checks subscription status, open pause, trigger, role, and reason eligibility.
resolve-duration-mode	Resolves fixed-day, scheduled-time, or open-ended pause and validates bounds.
compute-pause-fee-applicability	Decides whether the optional payment gate runs.

Example rule intent:

```
rule "WIK pause requires active subscription"
when
    $req: PauseRequestFact()
    $sub: SubscriptionStateFact(subscriptionId == $req.subscriptionId, statusCode != "ACTIVE")
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("INVALID_STATE_FOR_PAUSE")
    }
end

rule "WIK customer self-service cannot use admin-only reason"
when
    $req: PauseRequestFact(initiatingActorRole == "CUSTOMER_SELF_SERVICE", eligible == null)
    $reason: PauseReasonFact(code == $req.pauseReasonCode, adminOnly == true)
then
    modify($req) {
        setEligible(false);
        setEligibilityReasonCode("ADMIN_ONLY_PAUSE_REASON")
    }
end

rule "WIK fixed days duration within bounds"
when
    $req: PauseRequestFact(durationDays != null, durationValid == null)
    $reason: PauseReasonFact(code == $req.pauseReasonCode)
    $cfg: OperatorConfigFact()
    eval($req.getDurationDays() * 24 >= $cfg.getMinPauseHours()
        && ($reason.getMaxDurationDays() == null || $req.getDurationDays() <= $reason.getMaxDurationDays())
        && $req.getDurationDays() <= $cfg.getMaxFutureScheduledResumeDays())
then
    java.time.Instant resumeAt = java.time.Instant.now().plusSeconds($req.getDurationDays() * 86400L);
    modify($req) {
        setDurationMode("FIXED_DAYS");
        setResumeScheduledAt(resumeAt.toString());
        setDurationValid(true)
    }
end
```

10. Compensation Paths

Failure point	BPMN compensation
Drools rejects request	Emit SubscriptionPauseRejected; finalize FAILED.
Pending status update fails	No committed mutation expected; emit rejected; finalize FAILED.
Billing intent fails	Run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
Payment timeout	Run bil01-revert-pending-charge; run sub-lm-revert-to-prior; emit rejected; finalize FAILED.

Failure point	BPMN compensation
Fulfillment fails	Revert pending charge if any; run sub-lm-revert-to-prior; emit rejected; finalize FAILED.
Clear restrictions fails	Route to manual recovery; network may already be suspended.
Pause-history write or state commit fails	Route to manual recovery.
Kafka emit fails	Outbox retries. BPMN should not mark success event delivered synchronously.
Notification fails	Log and continue. Notification is reliability-isolated.

11. Worked Example

Kamau calls the WIK call center and asks to pause fiber service for 31 days while travelling.

Request:

```
POST /api/subscriptions/sub_01HZ_KAMAU_FIBER/pause
Idempotency-Key: pause-sub_01HZ_KAMAU-vacation-20260315
Content-Type: application/json

{
  "operatorCode": "WIK",
  "pauseTrigger": "CUSTOMER_REQUESTED_PAUSE",
  "pauseReasonCode": "CUSTOMER_VACATION",
  "durationDays": 31,
  "notes": "Customer travelling to Mombasa for holiday season",
  "initiatingWorkflowKind": "BO_UI",
  "initiatingWorkflowRef": "cc-ticket-2026-03-15-3301"
}
```

Runtime result:

```
API validates request
-> loads SUB-LM subscription summary: ACTIVE
-> loads WIK pause config and CUSTOMER_VACATION reason
-> Drools resolves FIXED_DAYS and resumeScheduledAt
-> framework creates operation subop_01HZ_KAMAU_PAUSE
-> Camunda starts sub-pause-wik
-> subscription moves ACTIVE -> PENDING_PAUSE
-> billing receives PAUSE_INTENT and returns no fee
-> FUL-04 suspends service
-> pause history row is inserted
-> subscription moves PENDING_PAUSE -> SUSPENDED
-> SubscriptionPaused event is emitted
-> notification is sent
-> operation is COMPLETED
```

After success:

- Subscription status is SUSPENDED.
- Customer service is suspended.
- subscription_pause_history.actual_resume_at remains NULL.
- Scheduler can later call RESUME with SCHEDULED_AUTO_RESUME when resume_scheduled_at arrives.

12. Cross-DD Dependencies

Dependency	Why it matters
DD_SUB-WF-PAUSE-01-Subscription_Pause-v1.0.md	Owns generic pause API, tables, events, and invariants.
DD_SUB-WF-FRAMEWORK-01-Subscription_Workflow_Framework-v1.0.md	Owns operation creation and process start.
DD_SUB-WF-FRAMEWORK-01-WORKERS-v1.0.md	Owns worker-topic contracts.
DD_SUB-LM-01-Subscription_Lifecycle_Management-v1.0.md	Owns subscription status source of truth.
BIL-01 and BIL-CFG-01	Own pause billing intent and any future pause fee.
FUL-04	Owns network suspension.
NOT-01	Owns customer notification.
SUB-WF-RESUME-01	Must close pause-history rows during manual or scheduled resume.
FOUNDATION_SCHEDULER	Must trigger scheduled auto-resume when configured.

13. Open Items

- WUG, WTZ, and YASSN pause flow satellites still require operator-specific deep dives.
- SCHEDULED_AUTO_RESUME must be supported by RESUME before fixed-duration pauses can resume automatically. Without it, operations can create scheduled pauses but operations staff must resume manually.